

- Standard Normal Vector  $Z$

= An  $n$  length random vector whose components  $Z_i$  are independent and standard normal.

$$E(Z) = 0$$

$$C_Z = \text{identity matrix}$$

$$f_Z(z) = \frac{1}{(2\pi)^{n/2}} \cdot e^{\{-\frac{1}{2}z^T z\}}$$

$$\left\{ \begin{array}{l} \text{PROOF: } f_Z(z_1, z_2, \dots, z_n) \\ \quad = f_{Z_1}(z_1) \cdot f_{Z_2}(z_2) \dots \\ \\ \quad = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}z_i^2} = \frac{1}{(2\pi)^{n/2}} \cdot e^{-1/2(z_1^2 + z_2^2 + \dots + z_n^2)} \\ \\ \quad = \frac{1}{(2\pi)^{n/2}} \cdot e^{-1/2(z^T z)} \end{array} \right\}$$

$$f_Z(z) = \frac{1}{\sqrt{(2\pi)^n |\det(I)|}} \cdot e^{-\frac{1}{2}z^T I^{-1} z} \quad \text{for standard normal}$$

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n |\det \Sigma|}} \cdot e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} \quad \begin{array}{l} \text{for multivariate gaussian} \\ (\Sigma = C_X = \text{cov. matrix of } X) \end{array}$$

Now suppose  $X = AZ + \mathcal{M}$ .

$$E(X) = A \cdot E(Z) + \mathcal{M} = \mathcal{M}$$

$$C_X = AC_Z A^T = AA^T$$

$$\begin{aligned} f_X(x) &= \frac{1}{|\det A|} \cdot f_Z(A^{-1}(x-\mathcal{M})) \\ &= \frac{1}{|\det A|} \cdot \frac{1}{(\sqrt{2\pi})^n} \cdot e^{-\frac{1}{2}(x-\mathcal{M})^T A^{-T} A^{-1}(x-\mathcal{M})} \\ &= \frac{1}{(2\pi)^{n/2} \sqrt{|\det(C_X)|}} \cdot e^{-\frac{1}{2}(x-\mathcal{M})^T (A^{-1})^T A^{-1}(x-\mathcal{M})} \quad (\Sigma = C_X) \end{aligned}$$

- $\bar{X}$  is multivariate Gaussian with mean vector  $\bar{\mu}$  and cov. matrix  $\Sigma$  ( $\bar{X} \sim N(\bar{\mu}, \Sigma)$ ) if for some  $A$  and  $\bar{\mathcal{M}}$ , it can be written as  $\bar{X} = A\bar{Z} + \bar{\mathcal{M}}$  where  $\bar{Z}$  is a standard normal vector and  $\Sigma = AA^T$ .

- Equivalent definitions of a Gaussian vector:

\*  $\bar{X} \sim N(\bar{\mu}, \Sigma)$  if for some  $A$  and  $\mathcal{M}$ , it can be written as  $X = AZ + \mathcal{M}$ .

\*  $\bar{X} \sim N(\bar{\mu}, \Sigma)$  if it has the pdf

$$f_X(x) = \frac{1}{(2\pi)^{n/2} \sqrt{|\det \Sigma|}} \cdot e^{-\frac{1}{2}\{(x-\mathcal{M})^T \Sigma^{-1}(x-\mathcal{M})\}}$$

\*  $\bar{X} \sim N(\bar{\mu}, \Sigma)$  iff for all vectors  $a \in \mathbb{R}^n$ , it turns out that  $a^T \bar{X}$  is univariate Gaussian  $N(a^T \bar{\mu}, a^T \Sigma a)$

- Affine transformations preserve Gaussianity.

Suppose  $X \sim N(\mu, \Sigma)$ .

Now consider  $Y = AX + b$ .

Then we have  $Y \sim N(A \cdot E[X] + b, A \Sigma A^T)$

$\Rightarrow Y$  is also a Gaussian vector.